

PROJECT TITAN

System Architecture Document

Educational Engineering Project

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IMPORTANT SCOPE NOTICE

Project TITAN is an educational design and analysis project. It is not a clinically validated device, regulatory submission, or product cleared for human use.

Document Control and Revision History

Revision	Date	Description	Prepared By	Status
A	July 2026	Initial system architecture baseline for the Project TITAN mechanical knee-support assembly.	Michael Lozada-Longog	Released

Source baseline: Project TITAN Engineering Design Report (July 2026), CAD assembly, completed engineering drawings, and Fusion 360 static FEA results documented by the author.

1. Purpose and Scope

This document defines the mechanical architecture, functional decomposition, component hierarchy, interfaces, load path, assembly sequence, and analytical boundaries of Project TITAN. It translates the requirements in TITAN-PRD-001 into a physical system definition suitable for CAD, FEA, manufacturing planning, and future prototype development.

Architecture Baseline

The current baseline is a nine-component structural assembly: main shaft, left and right side plates, upper and lower frames, left and right spacers, and two commercial bearings. No powered actuator, sensor, strap, human-interface pad, or clinically intended component is included in the released baseline.

2. System Mission

The system mission is to provide a modular mechanical platform that supports a bearing-guided rotational interface while maintaining structural alignment under a defined static load case. The architecture is intentionally simple so that structural behavior, manufacturing documentation, and design decisions can be evaluated clearly.

3. Functional Decomposition

Function ID	Primary Function	Allocated Elements	Verification
F-01	Carry external structural loads	Side plates, upper frame, lower frame	Static FEA and drawing review
F-02	Provide rotational support	Main shaft and two bearings	Assembly inspection
F-03	Maintain axial spacing and alignment	Left and right spacers	CAD dimensional review
F-04	Connect load-carrying members	Upper/lower frames and side plates	Assembly and interface review
F-05	Enable component replacement	Modular component architecture	BOM and assembly review
F-06	Provide analyzable geometry	Simplified parametric parts	FEA preprocessing and solve completion

4. Physical Architecture

The released design uses a dual-plate arrangement surrounding a central shaft and bearing stack. Upper and lower frame members establish the principal longitudinal structure. Spacers preserve axial separation and reduce direct side loading between adjacent components.

System hierarchy:

- Project TITAN Assembly
 - Structural Subsystem: left plate, right plate, upper frame, lower frame
 - Rotational Subsystem: main shaft, left bearing, right bearing
 - Alignment Subsystem: left spacer, right spacer
 - Documentation Subsystem: CAD, drawings, BOM, FEA, engineering report

5. Component Architecture and Responsibilities

Part No.	Component	Type	Primary Responsibility	Baseline Material
TITAN-001	Main Shaft	Custom	Defines the central rotational axis and transfers load into the bearing interfaces	6061-T6 aluminum in baseline documentation
TITAN-002	Left Side Plate	Custom	Carries structural load and supports one bearing seat	6061-T6 aluminum
TITAN-003	Right Side Plate	Custom	Provides opposing support and completes the dual-plate load path	6061-T6 aluminum
TITAN-004	Upper Frame	Custom	Provides the upper longitudinal structural connection and strap-slot concept geometry	6061-T6 aluminum
TITAN-005	Lower Frame	Custom	Provides the lower longitudinal structural connection and strap-slot concept geometry	6061-T6 aluminum
TITAN-006	Left Spacer	Custom	Maintains left-side axial clearance and alignment	6061-T6 aluminum
TITAN-007	Right Spacer	Custom	Maintains right-side axial clearance and alignment	6061-T6 aluminum
TITAN-008	Left Bearing	COTS concept	Provides rolling support at the left interface	52100 bearing steel
TITAN-009	Right Bearing	COTS concept	Provides rolling support at the right interface	52100 bearing steel

6. Mechanical Interfaces

Interface ID	Elements	Interface Type	Design Intent	Key Concern
IF-01	Main shaft ↔ bearing inner surfaces	Cylindrical load-transfer interface	Transfer radial bearing load while preserving axis alignment	Fit and contact definition
IF-02	Bearing outer surfaces ↔ side-plate bores	Bearing-seat interface	Transfer load into structural plates	Press-fit/retention not yet production-defined
IF-03	Spacers ↔ shaft/adjacent components	Axial spacing interface	Prevent component rubbing and control stack width	Tolerance stack-up
IF-04	Upper frame ↔ side plates	Rigid structural connection	Transfer upper-frame load into plate system	Fastener details remain conceptual
IF-05	Lower frame ↔ side plates	Rigid structural connection	Transfer lower-frame load into plate system	Fastener details remain conceptual

7. Load Path

The validated educational FEA load case applied bearing loads to the inner cylindrical bearing surfaces. The intended idealized load path is:

Load Path

Applied bearing load → bearing inner surface → bearing body/contact definition → side-plate bearing seat → side plates → upper/lower frame support regions → fixed structural boundary.

The final refined-mesh study reported a maximum von Mises stress of 54.95 MPa, minimum safety factor of 2.377, and maximum displacement of 0.004 in (approximately 0.10 mm). These values are accepted only for the documented idealized boundary conditions and do not constitute clinical validation.

8. Assembly Sequence

1. Prepare and inspect custom components and commercial bearings.
2. Seat the left and right bearings into their corresponding plate bores.
3. Position the main shaft through the bearing stack.
4. Install left and right spacers to establish the intended axial stack.
5. Align the second side plate with the shaft/bearing stack.
6. Attach the upper frame to the side-plate structure.
7. Attach the lower frame to the side-plate structure.
8. Perform interference, alignment, and free-rotation checks.
9. Record the as-built configuration and revision.

9. Analysis Architecture

The project used two levels of structural analysis: an isolated left-side-plate study for setup validation and a full-assembly static study for system-level evaluation. The full-assembly study required explicit contacts, constraints, loads, mesh generation, and cleanup of duplicate or unconstrained bodies.

Analysis Level	Purpose	Inputs	Outputs	Limitation
Component	Validate plate behavior and load application	6061-T6, fixed mounting holes, bearing load	Stress, displacement, safety factor	Does not capture assembly interactions
Full assembly	Evaluate integrated load path	All retained bodies, contacts, fixed supports, bearing loads	System stress, displacement, safety factor	Idealized contacts and supports

10. Architecture Constraints and Open Items

- The current design does not include a production-defined bearing retention shoulder, snap ring, adhesive, or interference fit.
- Fastener size, grade, thread engagement, torque, and locking method are not yet released.
- Human-contact surfaces, straps, padding, and anthropometric adjustability are outside the current baseline.
- The main shaft material is documented as aluminum in the current drawing package; a higher-strength shaft material remains a future trade-study item.
- Dynamic, fatigue, wear, sweat exposure, corrosion, and misuse conditions have not been experimentally validated.

11. Future Architecture Evolution

Version	Architecture Goal	Planned Additions
V2	Functional mechanical prototype	Production-like fits, retention, fasteners, improved geometry
V3	Instrumented biomedical concept	IMU, angle sensing, load sensing, data acquisition
V4	Connected rehabilitation platform	Bluetooth, dashboard, data logging
V5	Evaluation platform	Repeatability, usability, benchtop validation, human-factors planning

12. Conclusion

The Project TITAN architecture establishes a clear relationship between requirements, physical components, interfaces, and verification evidence. The current system is a mechanically focused educational baseline. Its principal strengths are modularity, analyzability, manufacturability awareness, and traceable documentation; its principal maturity gaps are production hardware definition, physical prototyping, and experimental validation.

References

Project TITAN Engineering Design Report, Michael Lozada-Longog, July 2026.

TITAN-PRD-001, Product Requirements Document, Revision B.

Project TITAN CAD assembly and Fusion 360 simulation outputs, July 2026.